MILITARY SPECIFICATION

INSERT, SCREW THREAD-LOCKED IN AND RING LOCKED, SERRATED
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for locked-in inserts. The method of locking in the insert is by means of a serrated collar and an accessory lockring with matching serrations installed within the parent material to prevent rotation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2024
QQ-A-250/6	Aluminum Alloy 5083, Plate and Sheet
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-H-1581	Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN:AMSMC-TDA-S(D), Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A.
APPROVED FOR PUBLIC RELEASE;
DISTRIBUTION UNLIMITED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-S-5626 - Steel; Chrome-Molybdenum (4140) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-I-6866 - Inspection, Penetrant Method of
MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-8837 - Coating, Cadmium (Vacuum Deposited)
MIL-S-8879 - Screw Threads; Controlled Radius Root with Increased Minor Diameter; General Specification for
MIL-L-8937 - Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-T-9047 - Titanium and Titanium Alloy Bars (Rolled or Forged) and Reforging Stock, Aircraft Quality

STANDARDS

FEDERAL

FED-STD-H28/2 - Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw Threads-UN and UNR Thread Forms
FED-STD-H28/20 - Screw Thread Standard for Federal Services. Section 20, Inspection Methods for Acceptability of UN, UNR, UNJ, M and MJ Screw-Threads

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1312 - Fasteners Test Methods

(See supplement 1 for list of MS sheet form standards.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents that are adopted by DoD shall be those in the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents which have not been adopted shall be the issue of the nongovernment documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

NATIONAL AEROSPACE STANDARDS (NAS)

NAS 672 - Plating - High Strength Steels - Cadmium

(Applications for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 DeSales Street, NW, Washington, DC 20036.)

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

AMS 2411 - Silver Plating for High Temperature Applications
AMS 4967 - Titanium Alloy Bars, Forgings and Rings
AMS 5731 - Steel Bars, Forgings, Tubing, and Rings, Corrosion and Heat Resistant
AMS 5734 - Steel Bars, Forgings, Tubing, Corrosion and Heat Resistant
AMS 6382 - Steel Bars, Forgings and Rings

(Applications for copies should be addressed to the Society of Automotive Engineers Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 108 - Steel Bars, Carbon, Cold-Finished, Standard Quality
ASTM A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment and Systems

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)
(Nongovernment documents are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 MS sheets. The individual item requirements shall be as specified herein and in accordance with the applicable MS sheets. In the event of any conflict between requirements of this specification and the MS sheet, the latter shall govern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.2 Material. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable. The insert shall be a one piece all metal unit. The insert and lockring shall be made from the following materials as specified in the applicable MS sheet (see 3.1) when inspected in conformance with 4.6.2.

3.2.1 Carbon and alloy steel. Carbon steel shall be grade 1117 (UNS G11170) in accordance with ASTM A 108. Alloy steel shall be grade 4140 (UNS G41400) in accordance with MIL-S-5626 or AMS 6382.

3.2.2 Corrosion-resistant steel. Corrosion-resistant steel shall be type A286 (UNS S66286) in accordance with AMS 5731 or AMS 5734.

3.2.3 Titanium alloy. Titanium alloy shall be type Ti-6Al-4V (UNS R56400) in accordance with MIL-T-9047 or AMS 4967.

3.3 Protective finish and surface treatment. The insert and lockring shall be furnished with a protective plating or surface treatment as specified in the applicable MS sheet (see 3.1) when inspected in conformance with 4.6.3.

3.3.1 Carbon and alloy steel. Inserts of carbon or alloy steel and lockrings of carbon steel shall be cadmium plated in accordance with QQ-P-416, Type II Class 3.

3.3.2 Corrosion-resistant steel. Corrosion-resistant inserts and lockrings shall be cleaned, descaled and passivated in accordance with ASTM A 380. When specified in the applicable MS sheet (see 3.1), corrosion-resistant steel inserts shall be silver plated in accordance with AMS 2411, .0002 inch minimum thickness. When specified in the applicable MS sheet (see 3.1), corrosion-resistant steel inserts shall have a dry film lubricant applied in accordance with MIL-L-8937.

3.3.3 Titanium alloy. Titanium alloy inserts shall have an anodic coating.

3.4 Design, dimensions and tolerances. Design, dimensions and tolerances shall be in accordance with the requirements of the applicable MS sheet (see 3.1) and shall apply after application of the protective finish or surface treatment specified in 3.3.

3.4.1 Threads. Threads shall be right hand in accordance with the applicable MS sheet (see 3.1)

3.4.1.1 External threads. External threads shall be in accordance with FED-STD-H28/2 with pitch and minor diameters as specified in the applicable MS sheet (see 3.1) for fine and coarse threads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.4.1.2 Internal threads. Internal threads shall be in accordance with MIL-S-8879 as specified in the applicable MS sheet (see 3.1). Internal threads which include locking feature shall be displaced or deformed as specified in 3.4.2. The insert without lubricant shall allow the "go" gage to enter not less than one-half turn before engagement of locking feature. When an insert with lubricant prevents the use of standard gages, the insert shall permit a minimum free rotational bolt thread engagement of three-fourths turn before engagement of the locking feature.

3.4.1.3 Thread forming. Threads may be produced either by machining, grinding, or fully formed by a single rolling process. In the event the external threads are rolled, the grain flow in the threads shall follow the general thread contour with the maximum density at the bottom of the root radius as shown in figure 1 in accordance with 4.7.6.

3.4.1.4 Incomplete and lead threads. Incomplete and lead threads, unless otherwise specified shall be in accordance with MIL-S-8879.

3.4.1.5 Thread concentricity. The internal thread pitch diameter shall be concentric with the external thread pitch diameter within .006 FIM.

3.4.2 Internal thread locking feature. The internal thread of the insert shall be distorted to impose friction between the insert and the inserted item. The feature shall not operate by means of separate movement from the installation, nor depend upon axial load on the insert wall.

3.4.2.1 Self-locking torque. When the insert is properly installed in the parent material, the locking feature shall provide a locking torque with a mating screw or bolt that is within the limits specified in table I when tested in accordance with 4.7.2. Inserts shall not be subjected to self-locking tests after testing as specified in 4.7.3, 4.7.4 or 4.7.5

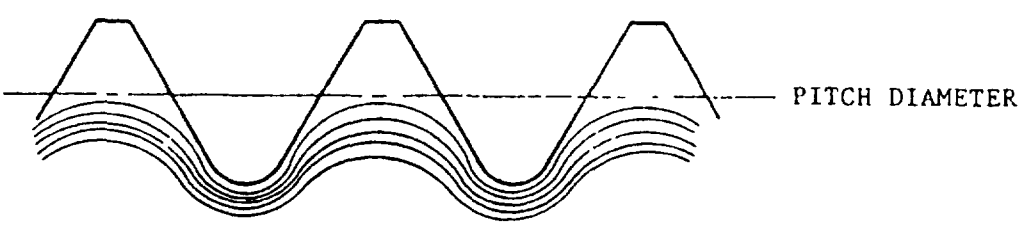


Figure 1. Thread grain flow.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE I. Internal thread, self-locking torque.

INSERT INTERNAL THREAD NOMINAL SIZE FINE OR COARSE	MS51498, MS51991, MS51993	
	MAX LOCKING TORQUE INCH-LBS	MIN BREAKAWAY TORQUE INCH-LBS
.112	5	1/
.138	10	1.0
.164	15	1.5
.190	18	2.0
.250	30	3.5
.3125	60	6.5
.375	80	9.5
.4375	100	14.0
.500	150	18.0
.5625	200	24.0
.625	300	32.0
.750	400	50.0
.875	600	70.0
1.000	800	90.0

1/ Some indication of torque

3.4.3 Locking serrations.

3.4.3.1 Insert locking serrations. Insert locking serration dimensions shall conform in size to MS51498, MS51991 or MS51993 and applicable dash number and shall be of full form.

3.4.3.2 Lockring serrations. Lockring serration dimensions for both internal and external serrations shall conform in size to MS51990 or MS51997 and applicable dash number.

3.4.3.3 Serration concentricity. The internal serrations shall be concentric with the external serrations within .006 FIM.

3.4.4 Surface texture. Surface texture of the insert and lockring prior to plating shall not exceed the values specified in the applicable MS sheet (see 3.1) and shall be in accordance with ANSI B46.1 when tested in conformance with 4.6.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.5 Mechanical properties. Inserts conforming to the design and dimensions specified in the applicable MS sheet (see 3.1) and having load ratings as specified in table II shall be capable of developing a minimum tensile strength, a minimum proof strength and have shear engagement areas in accordance with table II.

TABLE II. Mechanical Properties.

LOAD RATING KSI	MINIMUM TENSILE STRENGTH KSI	MINIMUM PROOF STRENGTH KSI	SHEAR ENGAGEMENT AREA (EXTERNAL THREADS)		
			MS51991		
			-100 NUMBERS	-200 NUMBERS	-300 NUMBERS
125	125	125	TABLE V COL 1	TABLE V COL 1	TABLE V COL 2

LOAD RATING KSI	MINIMUM TENSILE STRENGTH KSI	MINIMUM PROOF STRENGTH KSI	SHEAR ENGAGEMENT AREA (EXTERNAL THREADS)		
			MS51993		
			-500 NUMBERS	-640 NUMBERS	-800 NUMBERS
220	240	220	TABLE V COL 3	TABLE V COL 4	TABLE V COL 5

LOAD RATING KSI	MINIMUM TENSILE STRENGTH KSI	MINIMUM PROOF STRENGTH KSI	SHEAR ENGAGEMENT AREA (EXTERNAL THREADS)		
			MS51498		
			-500 NUMBERS	-640 NUMBERS	-800 NUMBERS
220	240	220	TABLE V COL 6	TABLE V COL 7	TABLE V COL 8

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.5.1 Minimum applied axial load. The installed insert (with lockring in place) shall develop the minimum axial load as specified in table III. Axial values are based on the tensile stress area of the applicable test bolt and the minimum tensile strength specified in table II when tested as specified in 4.7.3.

TABLE III. Axial load at specified stress level.

NOMINAL SIZE (INTERNAL THREAD)	TENSILE 1/ STRESS AREA (IN^2) OF BOLT		MINIMUM APPLIED AXIAL LOAD (LBS)		
			MS51991 125 KSI 2/		MS51498 & MS51993 240 KSI 3/
	COARSE THREAD	FINE THREAD	COARSE THREAD (UNJC)	FINE THREAD (UNJF)	FINE THREAD (UNJF)
.112	.0060	.0066	750	825	----
.138	.0091	.0101	1,140	1,260	----
.164	.0140	.0147	1,750	1,840	----
.190	.0175	.0200	2,190	2,500	4,800
.250	.0318	.0364	4,000	4,550	8,750
.3125	.0524	.0580	6,600	7,300	13,900
.375	.0775	.0878	9,700	11,000	21,000
.4375	.1063	.1187	13,300	14,800	28,500
.500	.1419	.1599	17,700	20,000	38,400
.5625	---	.203	----	25,400	----
.625	---	.256	----	32,000	----
.750	---	.373	----	46,600	----
.875	---	.509	----	63,600	----
1.000	---	.663	----	82,900	----

1/ The tensile stress areas used for the calculation of the axial load values are based on the stress area per formula specified in FED-STD-H28/2.

2/ The minimum applied axial loads shown are the product of the applicable tensile stress area times 125 KSI.

3/ The minimum applied axial loads shown are the product of the tensile stress area times 240 KSI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.5.2 Proof load. The installed insert shall be capable of withstanding proof loads as specified in table IV when tested in conformance with 4.7.3.

TABLE IV. Proof loads for inserts internal threads.

NOMINAL SIZE (INTERNAL THREAD)	TENSILE 1/ STRESS AREA (IN^2) OF BOLT		MINIMUM APPLIED PROOF LOAD (LBS)		
			MS51991 125 KSI 2/		MS51498 & MS51993 220 KSI 3/
	COARSE THREAD	FINE THREAD	COARSE THREAD (UNJC)	FINE THREAD (UNJF)	FINE THREAD (UNJF)
.112	.0060	.0066	750	825	----
.138	.0091	.0101	1,140	1,260	----
.164	.0140	.0147	1,750	1,840	----
.190	.0175	.0200	2,190	2,500	4,400
.250	.0318	.0364	4,000	4,550	8,000
.3125	.0524	.0580	6,600	7,300	12,800
.375	.0775	.0878	9,700	11,000	19,300
.4375	.1063	.1187	13,300	14,800	26,100
.500	.1419	.1599	17,700	20,000	35,200
.5625	---	.203	----	25,400	----
.625	---	.256	----	32,000	----
.750	---	.373	----	46,600	----
.875	---	.509	----	63,600	----
1.000	---	.663	----	82,900	----

1/ The tensile stress areas used for the calculation of the axial load values are based on the stress area per formula specified in FED-STD-H28/2.

2/ The minimum applied axial loads shown are the product of the applicable tensile stress area times 125 KSI.

3/ The minimum applied proof loads shown are the product of the tensile stress area times 220 KSI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.5.3 Resistance to pullout. The insert shall demonstrate minimum shear engagement areas as specified in table V. The installed insert with lockring in place as specified in 4.1.3 shall have a minimum resistance to pullout in accordance with table VI. Values specified in table VI are based on the minimum shear engagement areas of the insert (Table V) when installed in a test block of 25 KSI shear strength when tested as specified in 4.7.4.

TABLE V. Minimum shear engagement area of insert external thread 1/

NOMINAL SIZE (INTERNAL THREAD)	MINIMUM SHEAR ENGAGEMENT AREA (IN^2) - INSERT, NON LOCKED 1/							
	MS51991		MS51993			MS51498		
	-100 & -200 NUMBER	-300 NUMBERS	-500 NUMBERS	-640 NUMBERS	-800 NUMBERS	-500 NUMBERS	-640 NUMBERS	-800 NUMBERS
	COL 1 COARSE THREAD (NS)	COL 2 FINE THREAD (NS)	COL 3 FINE THREAD (NS)	COL 4 FINE THREAD (NS)	COL 5 COARSE THREAD (NS)	COL 6 FINE THREAD (NS)	COL 7 FINE THREAD (NS)	COL 8 FINE THREAD (NS)
.112	.0312	----	----	----	----	----	----	----
.138	.0576	----	----	----	----	----	----	----
.164	.0976	----	----	----	----	----	----	----
.190	.1442	----	.1500	.1917	.2516	.1543	.2047	.2965
.250	.2034	----	.2788	.3550	.4484	.2917	.3790	.5265
.3125	.2663	----	.4521	.5678	.7165	5546	.6987	.9281
.375	.3557	----	.6605	.8398	1.0690	.7826	.9992	1.2547
.4375	.4708	----	.8944	1.1340	1.4417	.9124	1.1938	1.6272
.500	.5872	----	1.1938	1.5126	1.9146	1.5757	1.7024	2.2785
.5625	---	.9983	----	----	----	----	----	----
.625	---	1.4120	----	----	----	----	----	----
.750	---	1.6990	----	----	----	----	----	----
875	---	2.2641	----	----	----	----	----	----
1.000	---	2.8387	----	----	----	----	----	----
	MINIMUM SHEAR ENGAGEMENT AREA (IN^2) - INSERT, LOCKED 1/							
.112	.0270	----	----	----	----	----	----	----
.138	0508	----	----	----	----	----	----	----
.164	.0941	----	----	----	----	----	----	----
.190	.1340	----	.1384	.1769	.2323	1461	.1960	.2828
.250	.1896	----	.2607	3319	4249	.2669	.3542	.5017
.3125	.2547	----	.4146	.5303	.6790	.5212	.6653	8947
.375	.3403	----	.6191	.7984	1.0276	.7436	9602	1 2157
.4375	.4534	----	.8445	1.0831	1 3908	.8710	1.1524	1.5858
.500	.5678	----	1.1470	1.4658	1.8678	1.5254	1.6701	2.2282
.5625	---	.9748	----	----	----	----	----	----
.625	---	1.3766	----	----	----	----	----	----
.750	---	1.6623	----	----	----	----	----	----
.875	---	2.2283	----	----	----	----	----	----
1.000	---	2.7769	----	----	----	----	----	----

1/ Shear engagement area is the assembled dimensional value for the overall engaged area of mating thread members. It does not represent a dimension of either of the members in an unassembled condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE VI. Resistance to pullout, insert external threads

NOMINAL SIZE (INTERNAL THREAD)	MINIMUM RESISTANCE TO PULLOUT (LBS) - INSERT, NON LOCKED 1/							
	MS51991		MS51993			MS51498		
	-100 & -200 NUMBERS COARSE THREAD (NS)	-300 NUMBERS FINE THREAD (NS)	-500 NUMBERS FINE THREAD (NS)	-640 NUMBERS FINE THREAD (NS)	-800 NUMBERS COARSE THREAD (NS)	-500 NUMBERS FINE THREAD (NS)	-640 NUMBERS FINE THREAD (NS)	-800 NUMBERS FINE THREAD (NS)
.112	780	----	----	----	----	----	----	----
.138	1,420	----	----	----	----	----	----	----
.164	2,440	----	----	----	----	----	----	----
.190	3,600	----	3,750	4,790	6,290	3,860	5,120	7,410
.250	5,080	----	6,970	8,880	11,210	7,290	9,480	13,160
3125	6,660	----	11,300	14,200	17,910	13,870	17,470	23,200
.375	8,890	----	16,510	21,000	26,720	19,560	24,980	31,370
.4375	11,770	----	22,360	28,350	36,040	22,800	29,850	40,680
.500	14,680	----	29,850	37,810	47,870	39,390	42,560	56,960
.5625	----	24,960	----	----	----	----	----	----
.625	----	35,300	----	----	----	----	----	----
.750	----	42,470	----	----	----	----	----	----
.875	----	56,600	----	----	----	----	----	----
1.000	----	70,970	----	----	----	----	----	----
	MINIMUM RESISTANCE TO PULLOUT (LBS) - INSERT, LOCKED 1/							
.112	675	----	----	----	----	----	----	----
.138	1,270	----	----	----	----	----	----	----
.164	2,350	----	----	----	----	----	----	----
.190	3,350	----	3,460	4,420	5,810	3,650	4,900	7,070
.250	4,740	----	6,520	8,300	10,620	6,670	8,860	12,540
.3125	6,370	----	10,360	13,260	16,980	13,030	16,630	22,370
.375	8,510	----	15,480	19,960	25,690	18,590	24,000	30,390
.4375	11,340	----	21,110	27,080	34,770	21,780	28,810	39,640
.500	14,200	----	28,680	36,640	46,700	38,140	41,750	55,700
.5625	----	24,370	----	----	----	----	----	----
.625	----	34,420	----	----	----	----	----	----
750	----	41,560	----	----	----	----	----	----
.875	----	55,710	----	----	----	----	----	----
1.000	----	69,420	----	----	----	----	----	----

1/ Installed in a test block with a shear strength as specified in 3.5.3

3.5.4 Rotational resistance. The insert shall produce a minimum torque resistance value not less than that specified in table VII when installed in conformance with MS51995 and the applicable MS sheet (see 3.1) and tested in accordance with 4.7.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE VII. Rotational resistance strength.

INTERNAL THREAD SIZE	MINIMUM ROTATIONAL RESISTANCE (TORQUE-OUT IN INCH LBS)		
	MS51991	MS51993	MS51498
.112	20	---	---
.138	30	---	---
.164	45	---	---
.190	65	120	400
.250	150	200	650
.3125	300	400	900
.375	550	650	1,300
.4375	850	900	2,300
.500	1,100	1,300	3,000
.5625	1,600	---	---
.625	2,300	---	---
.750	4,000	---	---
.875	6,000	---	---
1.000	8,000	---	---

3.5.4.1 Removal and replacement. The insert and lockring shall meet the requirements of 3.5.4 when replacing an identical insert and lockring (see 3.1) in an existing hole conforming to MS51994 and MS51996 and installed as specified in MS51995 provided the hole is undamaged by the insert removal operation.

3.5.5 Hardness.

3.5.5.1 Insert hardness. The insert shall meet the hardness as specified in the applicable MS sheet (see 3.1) when tested in accordance with 4.7.1.

3.5.5.2 Lockring hardness. The lockring of carbon steel shall be case hardened to the depth of .001 to .004 inch and be capable of developing a hardness as specified in the applicable MS sheet (see 3.1) when tested in accordance with 4.7.1. Corrosion-resistant steel lockrings shall not be case hardened.

3.6 Metallurgical properties.

3.6.1 Discontinuities. The insert and lockring shall not exhibit discontinuities exceeding the following limitations when tested in accordance with 4.6.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.1 Cracks. (see 6.3.1) The insert and lockring shall be free of cracks in any direction or location.

3.6.1.2 Laps and seams. (see 6.3.2 and 6.3.3) The insert and lockring may possess laps and seams except in locations specified in 3.6.2. Discontinuity depths shall not exceed 20 percent of the thread height (sharp V thread) for the insert and mating lockring.

3.6.1.3 Inclusions. (see 6.3.4) The insert shall show no evidence of surface or subsurface inclusions at the thread root when examined in accordance with 4.6.7. Small inclusions in other parts of the insert or lockring not indicative of unsatisfactory quality, shall not be cause for rejection.

3 6.2 Thread discontinuities. (Laps, seams, and surface irregularities in rolled threads). Threads shall have no laps at the root or along the flanks as shown in figure 2. Multiple laps on sides of threads are not permissible regardless of location. A single lap is permissible along the side of the thread above the pitch diameter on either the pressure or non-pressure side (one lap per thread) provided it extends toward the crest and generally parallel to the side as shown in figure 3a. Crest craters, crest laps, or a crest lap in combination with a crest crater are permissible, provided the imperfection does not extend deeper than 20 percent of the basic thread height (see table VIII) as measured from the thread crest when the thread major diameter is at minimum size (See figure 3a). Slight deviation from the thread contour is permissible at the crest of the thread within the major diameter limits as shown in figure 3b. The incomplete thread at each end of the thread may also deviate slightly from contour.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

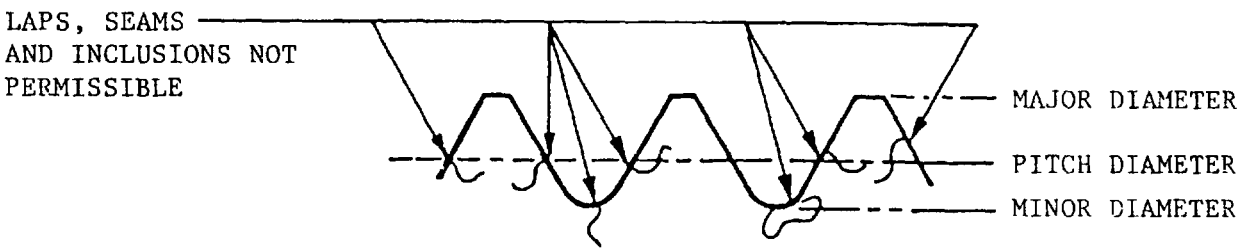


Figure 2. Laps, seams and surface thread.

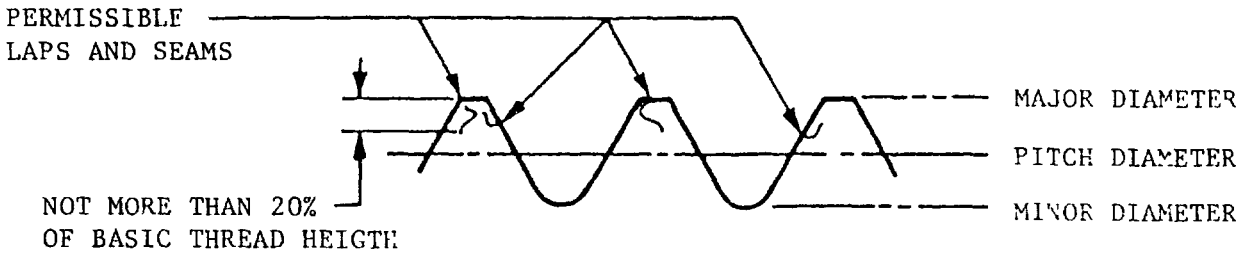


Figure 3a. Laps, seams and surface thread.

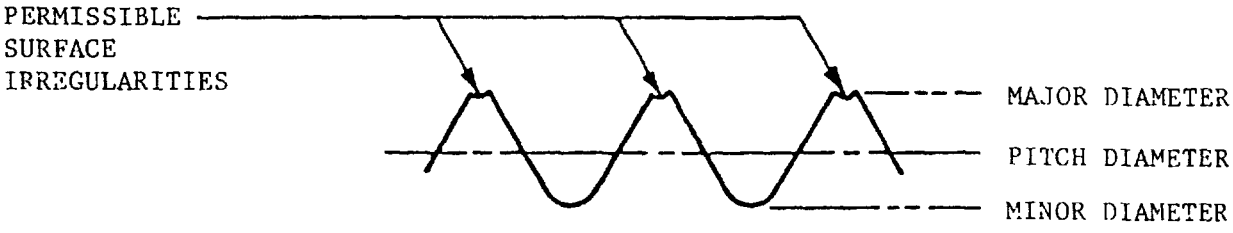


Figure 3b. Laps, seams and surface thread.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE VIII. Basic thread height, unified threads.

THREADS PER INCH	BASIC THREAD HEIGHT (REF)	20% BASIC THREAD HEIGHT
48	0.0135	0.0027
40	0.0162	0.0032
36	0.0180	0.0036
32	0.0203	0.0041
28	0.0232	0.0046
24	0.0271	0.0054
20	0.0325	0.0065
18	0.0361	0.0072
16	0.0406	0.0081
14	0.0464	0.0093
13	0.0500	0.0100
12	0.0541	0.0108
11	0.0590	0.0118
10	0.0650	0.0130

3.6 3 Grinding burns. The insert or lockring shall show no evidence of grinding burns.

3.7 Workmanship. Workmanship shall be consistent with the type of product, finish, and class of thread fit specified. The insert and lockring shall be of uniform quality free from laps, cracks, seams, inclusions, splits, or other defects or irregularities which would be detrimental to the performance of the insert or lockring in service use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test block fabrication. Test blocks shall be fabricated in accordance with figures 4 and 5 as applicable. Test blocks may exceed dimensions specified in figures 4 and 5 to accommodate multiple testing of inserts and lockrings except for tests of 4.7.3 and 4.7.4.

4.1.3 Test sample installation. Sample inserts and lockrings shall be installed in accordance with MS51995 in test blocks specified in 4.1.2.

4.1.3.1 Inspection of installed inserts and lockrings. The inserts and lockrings installed as specified in 4.1.3 shall be visually inspected under 10 diameters magnification. The presence of cracks in either test block, insert or lockring as a result of installation shall be cause for rejection. When visual inspection discloses a condition which shows cause for further examination, the test specimen shall be penetrant inspected in accordance with MIL-I-6866.

4.1.3.2 Test bolts and screws. Bolts and screws for use in all tests shall have Class 3A threads in accordance with FED-STD-H28/2.

4.1.3.2.1 Bolts and screws used for testing as specified in 4.7.3, 4.7.4, and 4.7.5 shall be heat treated and plated as follows:

a. To test MS51991 inserts.
Heat treat test bolts and screws 160 KSI minimum.
Cadmium plate per QQ-P-416, Type II, Class 3.

b. To test MS51993 and MS51498 inserts.
Heat treat test bolts and screws 260 KSI minimum.
Cadmium plate per MIL-C-8837, Type I, Class 2 or NAS 672.

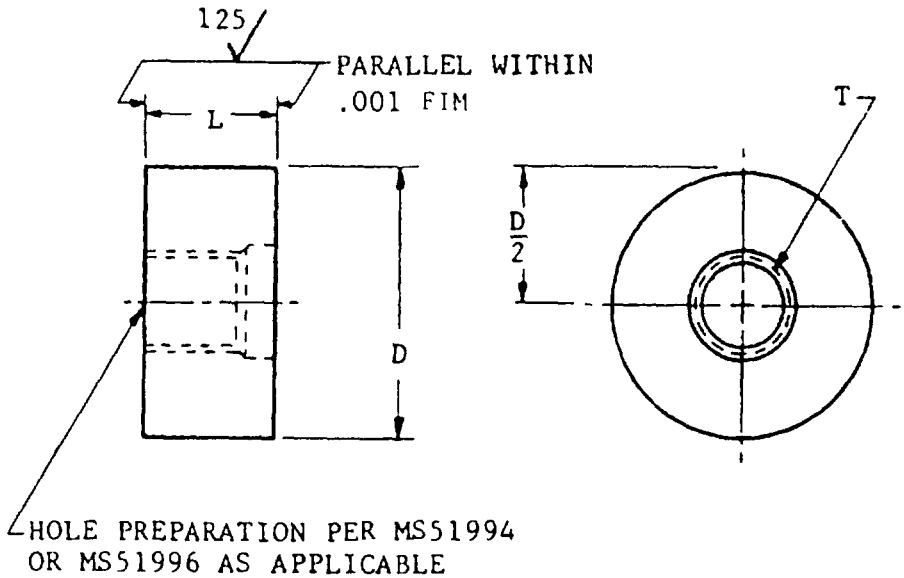
4.1.3.2.2 Bolts and screws used for internal thread self-locking tests of inserts as specified in 4.7.2 shall be heat treated as follows:

a. For MS51991 Carbon steel inserts use test bolts heat treated to 125 KSI minimum with cadmium plate per QQ-P-416, Type II, Class 3. For MS51991 corrosion-resistant steel inserts use corrosion-resistant steel bolts.

b. For MS51993 and MS51498 Alloy steel inserts use test bolts heat treated to 160 KSI minimum with cadmium plate per QQ-P-416, Type II, Class 3. For inserts of corrosion and heat resistant steel (A286) and silver plated use corrosion and heat resistant steel test bolts with minimum tensile strength of 130 KSI cleaned, descaled and passivated in accordance with ASTM A 380.

4.1.4 Test equipment and inspection facilities. The contractor shall insure that test and inspection facilities of sufficient accuracy; quality and quantity are established and maintained to permit performance of required inspections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.



1. DIMENSIONS
   T = NOMINAL THREAD DIAMETER OF APPLICABLE INSERT EXTERNAL THREAD.
   D = 4 X T (FOR T THREAD SIZE SMALLER THAN .500 DIA).
   3 X T (FOR T THREAD SIZES .500 DIA AND LARGER).
   L = LENGTH OF APPLICABLE INSERT PLUS .063.

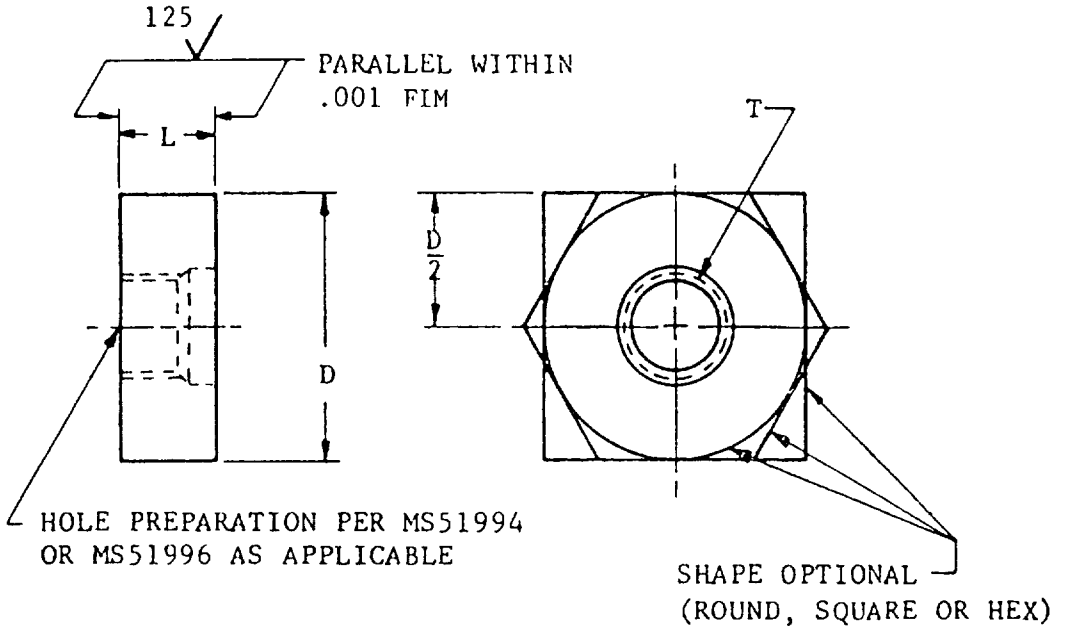
2. MATERIAL:
   (a) AXIAL TESTS (3.5.1 & 3.5.2) - ALUMINUM ALLOY 2024-T4 PER QQ-A-225/6 (BARS AND RODS)
   (b) PULLOUT TEST (3.5.3) - ALUMINUM ALLOY 5083-H321 PER QQ-A-250/6

3. ANODIZE PER MIL-A-8625, TYPE I, CLASS 1 OR CLASS 2.

4. PENETRANT INSPECT PER MIL-I-6866: DISCONTINUITIES OR CRACKS NOT ACCEPTABLE.

Figure 4. Axial load test blocks

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.



1. DIMENSIONS:
   T = NOMINAL THREAD DIAMETER OF APPLICABLE INSERT EXTERNAL THREAD.
   D = 4 X T (FOR T THREAD SIZES SMALLER THAN .500 DIA)
   3 X T (FOR T THREAD SIZES .500 DIA. AND LARGER)
   (a) L = LENGTH OF APPLICABLE INSERT MINUS .063 FOR ROTATIONAL RESISTANCE TESTS (3.5.4).
   (b) L = LENGTH OF APPLICABLE INSERT PLUS .063 FOR INTERNAL THREAD LOCKING FEATURE REQUIREMENT (3.4.2)

2. MATERIAL:
   ALUMINUM ALLOY, 2024-T4 PER QQ-A-225/6 (BAR).

3. ANODIZE PER MIL-A-8625, TYPE I, CLASS 1 OR CLASS 2.

4. PENETRANT INSPECT PER MIL-I-6866: DISCONTINUITIES OR CRACKS NOT ACCEPTABLE.

Figure 5. Torque test blocks

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table IX.

4.2.1 Inspection lot. An inspection lot shall consist of all inserts and lockrings covered by a single applicable MS sheet (see 3.1) produced under essentially the same conditions and offered for inspections at one time.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. A resubmitted lot shall be inspected using tightened inspection. Such lots shall be separated from new lots and clearly identified as reinspected lots.

4.3 Classification of defects. Classification of defects shall be in accordance with MIL-STD-105 and table X or XI as applicable.

4.3.1 Defect noncompliance. An insert or lockring exhibiting one or more defects shall be considered defective.

4.4 Inspection sampling. Inspection sampling shall be in accordance with MIL-STD-105 and the applicable inspection level and Acceptable Quality Level (AQL) specified in table X or XI as applicable.

4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with requirements of PPP-H-1581.

4.6 Method of inspection.

4.6.1 Visual and dimensional. The insert and lockring shall be examined to verify that physical dimensions, locking serrations, surface texture, grinding burns and workmanship are in accordance with the applicable requirements of 3.4, 3.4.4 and 3.7.

4.6.2 Material inspection. Material inspection shall consist of certification supporting verifying data that the materials used in fabricating the insert and lockring are in accordance with the applicable requirements of 3.2.

4.6.3 Treatment and finish inspection.

4.6.3.1 Protective finish inspection. Sample inserts and lockrings shall be inspected for minimum thickness and continuity of plating in accordance with QQ-P-416 or AMS2411 and requirements of 3 3.1, 3.3.2 and 3.3.3 as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Quality conformance inspection.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Group A		
Visual & Dimensional	3.4	4.6.1
Material	3.2	4.6.2
Protective Finish and surface treatment	3.3 (as applicable)	4.6.3 (as applicable)
Grinding burns	3.6.3	4.6.1
Workmanship	3.7	4.6.1
Packaging	5.1	4.5
Group B		
Threads	3.4.1 thru 3.4.1.4	4.6.4 & 4.7.6
Thread concentricity	3.4.1.5	4.6.4.3
Locking serrations	3.4.3	4.6.5
Surface texture	3.4.4	4.6.6
Group C		
Self locking torque	3.4.2.1	4.7.2
Minimum applied axial load	3.5.1	4.7.3
Proof load	3.5.2	4.7.3
Resistance to pullout	3.5.3	4.7.4
Rotational resistance	3.5.4	4.7.5
Hardness	3.5.5	4.7.1
Cracks	3.6.1.1	4.6.7
Laps and seams	3.6.1.2	4.6.7
Inclusions	3.6.1.3	4.6.7
Thread discontinuities	3.6.2	4.6.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE X. Classification of defects and inspection sampling for inserts.

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
MAJOR	AQL = 1.5 percent defective, Level II	
101	Design and dimensions incorrect (3.4)	Standard Measuring and Test Equipment (SMTE)
102	Threads not as specified (3.4.1 thru 3.4.1.4 as applicable)	SMTE
103	Internal threads not concentric with external threads (3 4 1.5)	SMTE
104	Self-locking feature missing, when required (3.4.2)	VISUAL
105	Surface texture (3.4.4)	SMTE
106	Cracks (3.6.1.1) and thread discontinuities (3.6.2)	SMTE
107	Imperfect insert serrations (3.4.3.1)	VISUAL
108	Treatment and finish (3.3 as applicable)	VISUAL
109	Mechanical properties (3.5 as applicable to inserts)	SMTE
MINOR	AQL = 2.5 percent defective, Level S1	
201	Overall length (3.4)	SMTE
202	Workmanship (3.7)	VISUAL

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

TABLE XI. Classification of defects and inspection sampling for lockring.

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
MAJOR	AQL = 1.5 percent defective, Level II	
101	Design and dimensions incorrect (3.4)	SMTE
102	Imperfect lockring serrations (3.4.3.2)	VISUAL
103	Concentricity between lockring internal and external serrations (3.4.3.3)	SMTE
104	Surface texture (3.4.4)	SMTE
MINOR	AQL = 4.0 percent defective	
201	Treatment and finish (3.3 as applicable)	VISUAL
202	Workmanship (3.7)	VISUAL

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

4.6.3.2 Cleaning, descaling and passivation inspection. Sample inserts and lockrings shall be inspected for cleaning, descaling and passivation in accordance with ASTM A 380 and requirements of 3.3.2.

4.6.4 Thread inspection.

4.6.4.1 External threads inspection. Sample inserts shall have their external threads inspected in accordance with FED-STD-H28/20, system 21, 3.4.1.1 and 3.4.1.3 requirements as applicable.

4.6.4.2 Internal threads inspection. Sample inserts shall have their internal threads inspected in accordance with FED-STD-H28/20, system 21, 3.4.1.2 and 3.4.1.3 requirements as applicable.

4.6.4.3 Thread concentricity inspection. Sample inserts shall have their internal thread pitch diameter concentric with the external thread pitch diameter in accordance with requirements of 3.4.1.5.

4 6.5 Locking serrations inspection. Sample insert and lockring locking serrations of both internal and external of lockring and external of insert shall be in accordance with requirements of 3.4.3 as applicable.

4.6.6 Surface texture inspection. Sample inserts and lockrings shall be inspected for surface texture with any of the surface examination and measurement methods specified in ANSI B46.1 in accordance with requirements of 3.4.4.

4.6.7 Metallurgical properties inspection. Sample inserts and lockrings shall be inspected for discontinuities in accordance with 3.6. Any crack detected during visual examination shall be cause for rejection of the lot. When visual evidence of discontinuities show cause for further inspection, sample inserts and lockrings shall be subjected to magnetic particle inspection in accordance with MIL-I-6868 for alloy steel, penetrant inspection in accordance with MIL-I-6866 for corrosion resistant steel and radiographic inspection in accordance with MIL-STD-453 for titanium alloy. Magnetic particle, penetrant or radiographic inspection alone shall not be cause for rejection. If indications are considered cause for rejection, representative samples shall be taken from those inserts and lockrings showing indications and these samples shall be further examined. Samples shall be sectioned and discontinuities measured under 10X magnification. The inspection shall be performed on finished inserts and lockrings free of lubrication and subsequent to any processing operation which could adversely affect the inserts and lockrings. Certification shall be required for identification marking.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

4.7 Mechanical properties inspection. In lieu of the mechanical properties sampling inspection of 4.7, mechanical properties inspection may consist of certification supporting verifying data that the insert and lockring have met the requirements of 3.5.

4.7.1 Hardness test. Sample inserts and lockrings shall be tested for hardness to meet the requirements of 3.5.5 in accordance with MIL-STD-1312, Test 6.

4.7.2 Self-locking torque test. Sample inserts shall be installed in test blocks in accordance with figure 5 and 4.1.3 to meet the requirements of 3.4.2.1. Bolts and screws shall be in accordance with 4.1.3.2 and shall have sufficient thread length to extend beyond the locking feature a minimum of two pitches (including thread chamfer). A new bolt or screw and a new sample insert shall be used for each complete fifteen cycle test. Bolts and screws shall assemble freely, with the fingers, up to the locking feature. The bolt or screw shall be engaged or disengaged from the assembled insert self-locking area for fifteen full installation and removal cycles without axial load on the insert. The test shall be run at a rate slow enough to yield a dependable measure of torque and to avoid heating of the bolt or screw. A bolt or screw shall be considered fully installed when two threads extend past the end of the insert locking feature; the removal cycle shall be considered complete when the locking feature is disengaged.

4.7.2.1 Maximum locking torque test. Maximum locking torque shall be the maximum torque value encountered on any installation or removal cycle and shall not exceed the applicable values specified in table I in accordance with the requirements of 3.4.2.1. Maximum locking torque readings shall be recorded on the first, seventh, and fifteenth installation cycle.

4.7.2.2 Minimum breakaway torque test. Minimum breakaway torque shall be the minimum torque required to start removal of the bolt or screw from a fixed position located between the first 1/4 turn to 1 turn of the removal cycle. Minimum breakaway torque readings shall be recorded at the start of the first, seventh and fifteenth removal cycle. The torque for any cycle shall not be less than the applicable values specified in table I in accordance with the requirements of 3.4.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

4.7.3 Axial strength (tensile and proof test). Sample inserts and lockrings installed in test blocks in accordance with figure 4 and 3.5.1 for tensile test and figure 4 and 3.5.2 for proof test as specified in 4.1.3 shall meet installation requirements specified in 4.1.3.1. The test bolt shall assemble freely into the insert prior to engaging the self-locking device, with finger torque. When testing MS51991 insert, the test bolt thread shall be of sufficient length to extend through the self-locking feature a minimum of two (2) pitches (including bolt thread chamfer). When testing MS51498 or MS51993 insert, the test bolt thread shall be of sufficient length to engage the full length of the internal thread. The bushing and test block clearance holes in the upper and lower yokes (see figure 6) shall have a free fit not in excess of 0.06 inch greater than the bushing and test block diameters. The test bolt clearance hole in the upper bushing (see figure 6) shall be 0.005 to 0.015 inch diameter larger than the nominal diameter of the test bolt. The bolt clearance hole in lower yoke shall have a diameter of 1½ x the nominal insert external thread diameter ±.015. The axial tensile load values determined by table II and specified in table III, as applicable, shall be applied to the test assembly and failure of the insert shall not occur below these loads. Proof test load values determined by table II and specified in table IV as applicable, shall be applied to the test assembly and test may be terminated when proof load is reached. In the event of test bolt failure by fracture below the applicable load values specified in table III or a test bolt failure by yielding (permanent set) below the applicable load values specified in table IV, the test shall be repeated until the applicable load value of the insert is reached or exceeded. Rate of loading shall not exceed 100 KSI per minute per square inch based on the shank diameter area of the bolt. Sample inserts shall meet the requirements of 3.5.1 and 3.5.2 as specified.

4.7.4 Resistance to pull-out test. Sample inserts and lockrings shall be installed in test blocks in accordance with figure 4 and 4.1.3 to meet the requirements of 3.5.3. The minimum single shear strength of the test block material shall be determined by means of double shear tests of coupons taken from the material from which the test blocks are fabricated. The procedure shall be the same as 4.7.3 except, to demonstrate the average shear engagement area, an axial load of sufficient magnitude to produce failure (pull-out) shall be applied to the assembly. The axial load test result shall be adjusted for test block shear strength as follows: The product of the axial load test result (pounds) and the ratio of the 25 KSI to the actual minimum shear strength shall be used as the resistance to pullout (pounds). The mean (arithmetical average) value as determined by five or more tests shall meet or exceed the values specified in table VI in accordance with 3.5.3. In the event of bolt failure below the applicable strength rating, the test shall be repeated until the pull-out strength of the insert is reached or exceeded. Rate of loading shall not exceed 100 KSI per minute based on the shank diameter area of the bolt. Whenever the pull-out values exceed the tensile strength of the test bolt, or the test equipment, it is permissible to use lower shear strength materials (15 KSI approx.) in lieu of specified test block materials. Double shear tests on coupons taken from the lower shear strength test blocks shall be performed and the resultant shear strength used to adjust the test results as stated above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

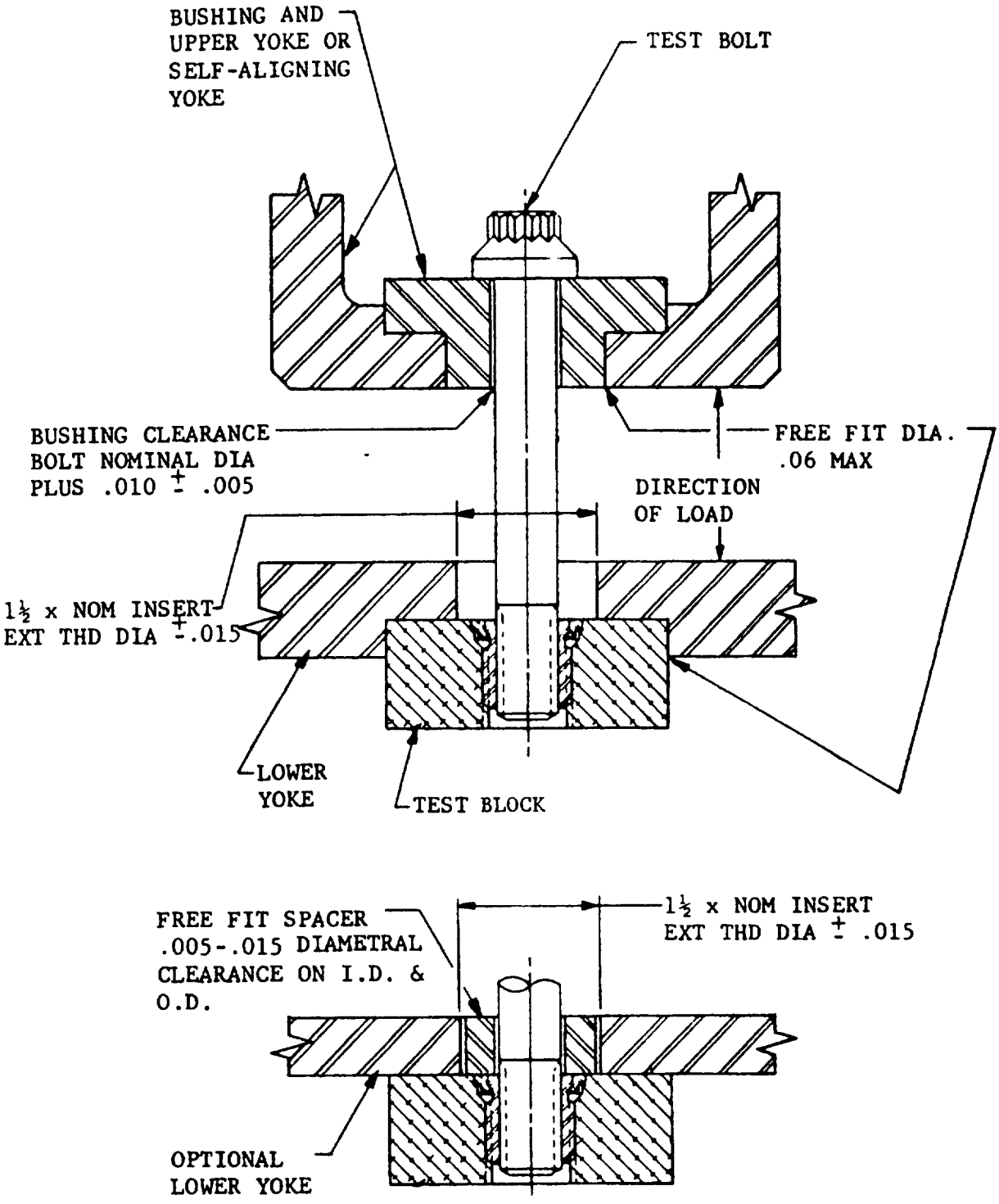
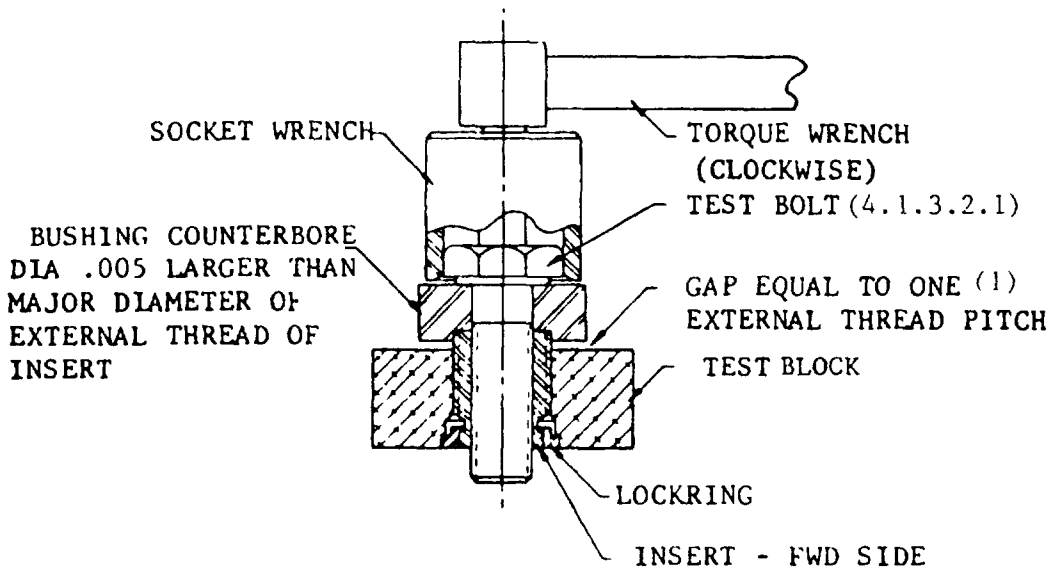
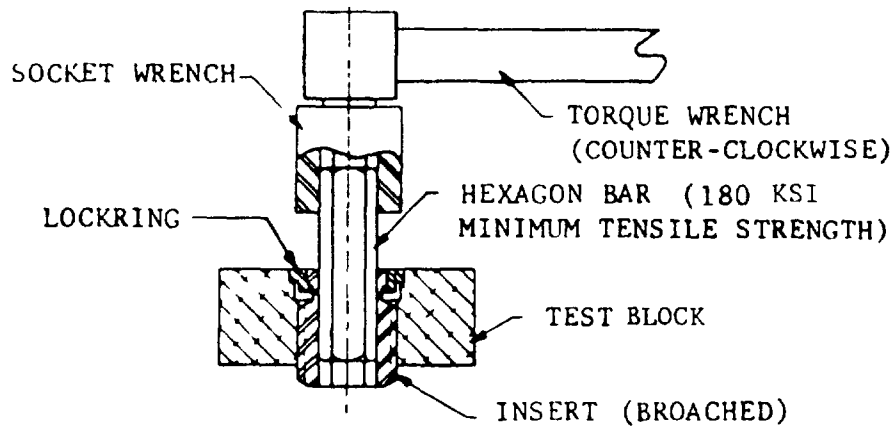


Figure 6. Axial strength and Resistance to pullout test fixtures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.



NOTE: ABOVE TEST FIXTURE USED WITH INSERT IN THE AS RECEIVED CONDITION



NOTE: ABOVE TEST FIXTURE USED WHEN PRODUCT IS INTERNALLY BROACHED WITH A HEXAGON WHOSE CROSS POINT DIMENSION DOES NOT EXCEED THE MAJOR DIAMETER OF THE INSERT INTERNAL THREAD. MORE ACCURATE RESULTS ARE OBTAINABLE WITH THIS METHOD AS THE INSERT WILL NOT TEND TO COLLAPSE OR TRANSMIT FRICTION INTO FINAL VALUES.

Figure 7. Rotational resistance test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

4.7.5 Rotational resistance test. Sample inserts and lockrings shall be installed in test blocks in accordance with figure 5 and 4.1.3 and utilizing a test assembly as illustrated in figure 7. With the forward side face of the insert installed in accordance with MS51995, the far side of the insert shall extend a distance approximately equal to one (1) external thread pitch past the far side of the test block allowing it to fit into the counterbore of the bushing. The insert shall fit into the bushing counterbore sufficiently close to prevent it from expanding under pressure. A test bolt or screw of 160 KSI minimum tensile strength as specified in 4.1.3.2.1 shall enter the insert from the back side (opposite normal entry). The test bolt or screw shall be torqued in a clockwise direction. Failure at applicable values below those specified in table VII in accordance with the requirements of 3.5.4 shall be cause for rejection.

4.7.6 Grain flow inspection. Sample inserts shall be inspected for grain flow for rolled threads by microexamination or macroexamination in accordance with the requirements of 3.4.1 3 when applicable. Specimens shall be taken from the finished insert as shown in figure 8. The inserts shall be etched in a suitable etchant for sufficient time to reveal the macrostructure property.

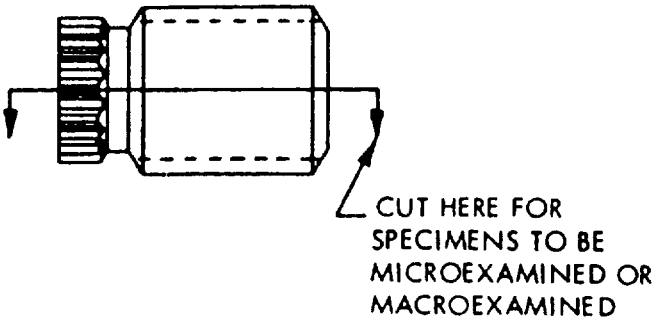


Figure 8 Metallurgical specimen.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Inserts and lockrings covered by this specification are intended as a general purpose fastener with a mechanical lock feature to resist rotation. See MS51998 for design and usage limitations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of applicable Military Standard (see 3.1).
c. Applicable Military Standard part number (see 3.1).
d. Level (degree) of protection, in accordance with PPP-H-1581, ordering data (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

6.3 Definitions.

6.3.1 Crack. A crack is a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

6.3.2 Lap. A lap is a surface defect appearing as a seam, caused by the folding over of metal fins or sharp corners and then rolling or forging them into the surface, but not welding them.

6.3.3 Seam. A seam is an unwelded fold or lap which appears as an opening in the raw material as received from the source.

6.3.4 Inclusions. Inclusions are non-metallic materials in a solid metallic matrix.

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Review Activities:
Army - AT, AV, ER, GL, MI
Air Force - 11, 82
NSA - NS
DLA - IS

User Activities:
Army - ME, GL
Navy - MC, OS

Preparing Activity:
Army - AR

(Project 5340-1586)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Munitions &
Chemical Command
ATTN: AMSMC-TDA-S(D)
Dover, NJ 07801-5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-45910C	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.

MILITARY SPECIFICATION

INSERT, SCREW THREAD-LOCKED IN AND RING LOCKED, SERRATED, GENERAL SPECIFICATION FOR

This supplement forms a part of Military Specification MIL-I-45910C, dated 11 March 1985.

MS SHEET FORM STANDARDS

MS51498 - Insert, Screw Thread - Locked in, Ring Locked, Serrated, High Strength, Oversize Replacer
MS51990 - Ring, Lock, Serrated
MS51991 - Insert, Screw Thread - Locked in, Ring Locked, Serrated
MS51993 - Insert, Screw Thread - Locked in, Ring Locked, Serrated, High Strength
MS51994 - Hole Preparation - Ring Locked Inserts and Studs, Standard Dimensions for
MS51995 - Fasteners, Ring Locked Inserts and Studs, Installation of
MS51996 - Hole Preparation - Ring Locked Inserts and Studs, High Strength, Standard Dimensions for
MS51997 - Ring, Lock, Serrated - High Strength

Preparing Activity:
Army - AR

FSC 5340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:10Z
Check the source to verify that this is the current version before use.